# Personality Classification from Online Handwritten Signature using k-Nearest Neighbor

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***Abstract*—Signature is the representation of a personal identity that written on a selected medium. Signature can describe people's personality, which includes character, potential, motivation, emotional stability, mental state, intellectual tendencies, and one's work habits. Understanding people's personality is useful to show personality traits in criminology, medical science, and counseling. Previous research has investigated the relationship between online signature and personality by using tablet digitizer. In this paper, we are improving the method by increasing the amount of respondents, using a classification algorithm, and using psychological tests for validation named Big Five Inventory Test (BFI Test). The two best features pressure and speed have been analyzed and classified using k-Nearest Neighbor (kNN). The highest classification accuracy from kNN is 87.5%. From 40 respondents who were involved, 90% were confirmed well with BFI test for the first or second dominant personality. As a conclusion, this research shows that online signature analysis could predict personality with high accuracy.**

***Keywords*— *Online Signature, Classification Algorithm, BFI Test, kNN***

## I. Introduction

Biometric system is a self-recognition technology that is using parts of human's body (such as fingerprint, face, and iris) or human behavior (such as voice, signature, and gait) [1], [2]. Among those biometric systems, signature is the most widely used and accepted for the biometric system in the community because of their ease of use [3], [4]. Many people are implementing a signature in various fields, such as finance and legal agreements [3]. A signature is a representation of a person's name or identity that is written on a selected medium [5], [6].

There are two types of signature, online and offline. Online signature is also called as dynamic signature, it is obtained through a digitizer that can provide data such as speed and pressure. An offline signature is also called a static signature obtained through stationery on paper media [7], [8], [9], [10].

In addition to verification, a signature is also used to determine personalities based on a measurable psychological analysis [11], [12]. Personality is an organized behavior from an individual [11]. A signature is also the brain-derived writings, resulting in thoughts, then reflected in behavior [11], [13]. Signature can describe the character, potential, motivation, emotional stability, mental state, intellectual tendencies, and one's work habits.

There are two ways to perform signature analysis by using general or measurement analysis. The general analysis consists of legible signatures, self-reliance, and negative strokes. Meanwhile, the measurement analysis performed a measurement of parameters including height, pressure, width, speed, and number of pen strokes [11].

This study uses a psychological test named BFI for validation purposes. The Big Five is a personality taxonomy that is built on a lexical approach, that is taken by grouping words or languages that are used in everyday life. It is used to describe the characteristics of a person or individual that distinguishes them from other individuals [11], [14]. It is also a grouping of thousands of traits into five large sets called personality dimensions. This grouping system is widely used in many published articles.

The five types of personality according to the BFI are Extraversion, Agreeableness, Conscientiousness, Neuroticism, and Openness [14]. Understanding people personality is useful to shows personality traits in criminology, medical science, and counseling [11].


Fig. 1. BFI Illustration, Verywellmind, Joshua Seong

In this research, the classification was performed using kNN. Classification is a data mining function which is useful for forecasting the class of unknown instances. kNN is a classification algorithm that is popular because it is simple, easy, and effective. kNN is also one of the top 10 algorithms used for data mining [15].

The objective of this research is to prove that signatures can be used to determine an individual's personality and this research also complements the previous study by increasing the amount of respondent, using classification algorithm, and using psychological tests for validation named BFI.

## II. RELATED WORKS

Several studies have discussed personality determination through online or offline signature analysis. In this chapter, we are presenting two similar studies for our guidelines in conducting this research.

### A. *Analysis of Signature for the Prediction of Personality Traits* [11]

Vaishali R. Lokhande and Bharti W. Gawali in 2017 performed a study by scanning and resizing signatures, then by doing preprocessing in the form of a grayscale image, binary image, and noise removal. The next process is to do image segmentation, then feature extraction, and finally the prediction of personality. The purpose of the study is to predict a person's personality including fear and honesty. Some of the features used to determine personality are the lines under the signature, the point at the end of the signature, the initial curve, the final stroke, and the dash. The amount of data taken is 60 from 10 people with six signatures. There are five categories of evaluations performed on signature scans with 500dpi resolution, such as the left, right, up, middle, and bottom categories. The artificial neural network algorithm and structural identification algorithms were yielding successive accuracy of 100%, 95%, 94%, 96%, and 92%. The test results were grouped according to the Big Five Personality criteria, the same as the one we used in our current study. The results of the study are expected to be used to identify personalities related to criminology, medical science, and counseling.

Although the results of the research generate high accuracy, determining personality using offline signatures requires relatively long steps as mentioned above, so determining personality through online signatures is more advantageous because it is faster.

### B. *Online Signature System Based on Pressure and Speed Features* [16]

Nor Azlin Rosli and Azlinah Hj Mohamed in 2012 performed an online signature analysis research using two features to determine a person's personality, namely pressure and speed. The sum of pressure was recognized based on every single pressure point that was obtained by the digitizer. Speed is obtained by dividing total number of distance with the time taken to makes a signature. The research is a development of previous research on the HoloCat Matrix. The research process includes data acquisition where the respondent will sign the digitizer tablet (Genius Mouse Pen 8 x 6) and take the x and y values. The research took five signatures for the test and six signatures for verification. The two tests performed produced good accuracy, and this was proven by comparing the results of the test with the results of the analysis performed by a graphologist manually.

According to the graphologist, even though the test results are accurate, the signature of the respondent on the digitizer tablet media does not reflect the signature written on paper media because the respondent must look at the monitor screen when they wrote their signature. This research has not conducted data classification and was not validated with manual psychological test.

## III. METHODOLOGY

The research methodology is presented in Fig. 2 which is consistent of Data Collection, Features Extraction, Classification, and Validation using BFI Test.

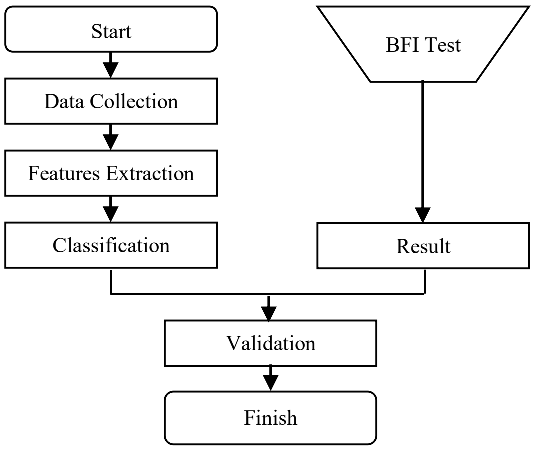


Fig. 2. Research Methodology

### A. *Data Collection*

This study is analyzing the online signature of 40 respondents from undergraduate and post-graduate college students. The first step is the respondents make one offline signature in the paper then the second step is to make three online signatures in a tablet digitizer called Wacom Tablet Signature Pad STU-540. The tablet specifications as follows: screen size 108 x 65 mm, pressure level 1024 (non-interpolated), and coordinate accuracy ±0.5 mm (center). By using the Signature Scope Application, the Wacom Signature Pad can obtain 62 types of a signature's biometric data, including number of down-strokes, total time from first down to the last pen-up, distance from start to end, average pen force in device units, pen speed. The last step is the respondents fill out the BFI test manually.

### B. *Features Extraction*

Six potential features that can be used for determining personality through online signature analysis are pressure, speed, width, height, baseline position, and number of strokes. To obtain features that produce the best accuracy, we can use the Eval Attribute Gain Info on The Waikato Environment for Knowledge Analysis (WEKA). In this study, the two best features that we used are pressure and speed. These two features are the same as in previous studies [16]. The shape of a signature that a person makes is never the same, but according to the data observation, the two chosen features tend to produce biometric data in the same range on the three data retrievals in the tablet digitizer. The pressure value is the average pen force in device units. The speed value is the root mean square (RMS) pen speed.

The range labeling of the selected features from 120 signatures is available in Table I.

TABLE I. RANGE LABELING

| Feature | Range | Label |
|---|---|---|
| Pressure | 389-585 | Light |
| | 586-782 | Medium |
| | 783-979 | Heavy |
| Speed | 38-120 | Slow |
| | 121-203 | Medium Speed |
| | 204-285 | Fast |

In this paper, we are labeling the features of pressure and speed into three ranges of values that refer to the previous study [16] by using our data. The way to determine the first, second, and third range of pressure is discussed below.

{The Highest Value (979) - The Lowest Value (389)} : 3. The result of the division is added to the smallest value until reaching the highest value (389+196=585, 586+196=782, 783+196=979). In the end, we can have the first range that is 389-585, the second is 586-782, and the third is 783-979. The determination of the speed range is done in the same way.

### C. Classification

The formula of the euclidean distance for defining the distance between two objects x and y in kNN is:

$$dxy = \sqrt{\sum_{i=1}^{n}(xi - yi)^2} \tag{1}$$

With:
*dxy* = euclidean distance
*xi* = object x to i
*yi* = object y to i
*n* = number of objects

The formula 1 is used to estimate the nearest neighbor of an object [15]. This research uses the kNN algorithm because of the relatively small amount of data and the need for simplicity.

WEKA 3.8 was used for classification processing in this study. It has been commonly used for machine learning and data mining that was initially evolved at The University of Waikato in New Zealand. WEKA consists of a collection of algorithms, and their visualization was used for data analysis and predictive modeling [17]. In WEKA, the kNN algorithm is included in the lazy classifier named IBk.

### D. BFI Test

Initially, the BFI test consists of 44 questions in English. Corresponding to the respondents' language preference, we changed the questions in Bahasa Indonesia. The result of the BFI Test was done by summing the values of each answer. The values are as follows: 1 (disagree strongly), 2 (disagree a little), 3 (neither agree nor disagree), 4 (agree a little), and 5 (agree strongly). The "Personality Test V2.8" application which is available for free on Google Play is used to determine the BFI test results. The application was chosen to obtain accurate results.

The details of the five types of personality according to the Big Five are presented in Table II.

TABLE II. THE BIG FIVE TYPES OF PERSONALITY [14]

| No. | Big Five | Explanation |
|---|---|---|
| 1 | Extraversion | Passion and enthusiasm (passionate about building relationships with other people), never hesitates to get acquainted and actively seek new friends, positive emotions, assertive (the ability to communicate feelings, thoughts, and desires honestly to others without harming or hurting others' feelings), and dares to say no if they disagree, so they tend to be able to become leaders. |
| 2 | Agreeableness | Sincere in sharing, feeling smooth, focusing on positive things in others, and in everyday life appears as a kind, collaborative and trustworthy individual. |
| 3 | Conscientiousness | Has the characteristic of being serious in performing tasks, being reliable, responsible, liking order and discipline, in daily life appears as a person who is timely, accomplished, meticulous, and likes to do work thoroughly. |
| 4 | Neuroticism | Tends to be anxious, easy to worry, tense, afraid, and easily nervous in facing problems that according to others are trivial, tends to be easily angry when dealing with situations that are not desired, and this type generally lacks tolerance to disappointment and conflict. |
| 5 | Openness | Has original openness of insight and ideas, ready to accept various inputs from various perspectives, and happy with a variety of new information, likes to learn new things, and likes to do new activities out of the ordinary. |

### E. Validation

Validation is the activity of checking or proving the validity or accuracy of something. In this research, we performed the validation by comparing the personality determination from signatures analysis and personality determination through the BFI test. The purpose of this validation is to prove that signature analysis can determine personality. Great validation values indicate that research has succeeded.

For validation purposes, data from the next two features were labeled based on BFI presented in Table III.

TABLE III. BFI LABELING

| Pressure | Speed | Personality based on [16] | Label According to BFI |
|---|---|---|---|
| Light | Slow | Sensitivity (High) & Rule-Consciousness (High) | Conscientiousness (3) |
| Light | Medium Speed | Sensitivity (High) & Apprehension (High) | Neuroticism (4) |
| Light | Fast | Sensitivity (High), Apprehension (Low), & Self-Reliance (High) | Openness (5) |
| Medium | Slow | Warmth (High), Emotional Stability (High), & Rule-Consciousness (High) | Agreeableness (2) |
| Medium | Medium Speed | Warmth (High), Emotional Stability (High), & Apprehension (High) | Agreeableness (2) |
| Medium | Fast | Warmth (High), Emotional Stability (High), Apprehension (Low), & Self-Reliance (High) | Agreeableness (2) |
| Heavy | Slow | Emotional Stability (Low), Dominance (High), Liveliness (High), & Rule-Consciousness (High) | Extraversion (1) |
| Heavy | Medium Speed | Emotional Stability (Low), Dominance (High), Liveliness (High), & Apprehension (High) | Extraversion (1) |
| Heavy | Fast | Emotional Stability (Low), Dominance (High), Liveliness (High), Apprehension (Low), & Self-Reliance (High) | Extraversion (1) |

Data labeling in Table III. shows the disproportionate division of BFI personality (Agreeableness and Extraversion appears three times), but the division has gone through the process of matching personality according to Cattell's 16 Personality Factor used by previous studies [16], [18] to the BFI test.

## IV. RESULT AND DISCUSSION

The following are the examples of respondents' offline signature, online signature, and pressure graph.

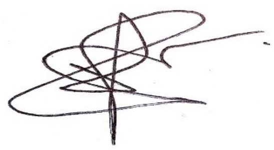

Fig. 3. The Example of Offline Signature (Subject R.12)

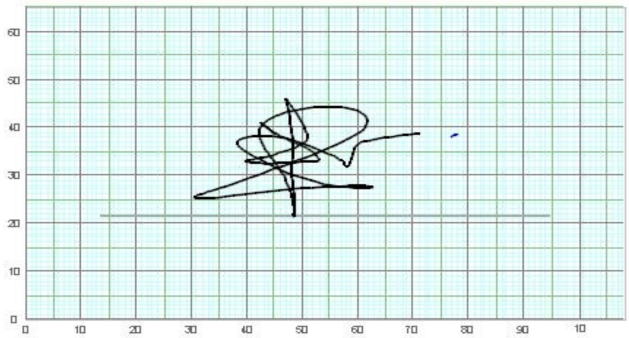

Fig. 4. The Example of Online Signature (Subject R.12)

The shape of our signature that we make is usually not exactly the same. The differences that generally occur are height, width, arch shape, and slope. The differences indicate that analyzing offline signatures tends to be more difficult than the online signature. In the other way, data from the online signature such as pressure and speed tend to be the in the same range as specified in Table I. For example, for Subject R.12, on the first until third online signatures, the pressure values are 764, 712, 661 and the speed values are 115, 109, 107. All the online signatures from subject R.12 show that the pressure value is in the range of medium and the speed value is in the range of slow. In the end, the two features (pressure and speed) that we used have shown obvious parameters for analysis purposes.

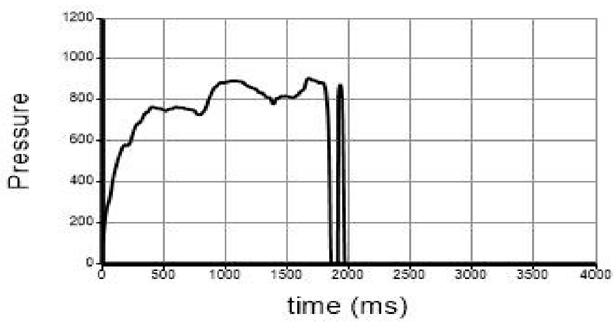


Fig. 5. The example of The Pressure Graph (Subject R.12)

From Fig. 5. we can see that there is two pen force in the total time of nearly two seconds and each pen force has a different pressure value. The pressure value is the average pen force in the digitizer.

The best data were taken from three signatures of each the respondent with the following details presented in Table IV.

TABLE IV. DATA RECAPITULATION

| Respondents | Pressure Value | Speed Value | Labeling Result | BFI Test | Match |
|---|---|---|---|---|---|
| R.01 | 726 | 51 | 2 | 2/1/3/5/4 | 1st |
| R.02 | 765 | 184 | 2 | 2/4/3/5/1 | 1st |
| R.03 | 773 | 117 | 2 | 1/2/5/3/4 | 2nd |
| R.04 | 615 | 126 | 2 | 2/5/1/3/4 | 1st |
| R.05 | 567 | 130 | 4 | 2/4/3/5/1 | 2nd |
| R.06 | 686 | 92 | 2 | 4/2/5/3/1 | 2nd |
| R.07 | 658 | 68 | 2 | 2/3/4/5/1 | 1st |
| R.08 | 396 | 40 | 3 | 2/1/3/5/4 | 3rd |
| R.09 | 768 | 89 | 2 | 2/1/5/3/4 | 1st |
| R.10 | 600 | 71 | 2 | 2/3/5/1/4 | 1st |
| R.11 | 653 | 163 | 2 | 2/3/5/1/4 | 1st |
| R.12 | 661 | 107 | 2 | 2/3/5/1/4 | 1st |
| R.13 | 755 | 136 | 2 | 2/3/4/5/1 | 1st |
| R.14 | 596 | 145 | 2 | 2/5/3/1/4 | 1st |
| R.15 | 727 | 126 | 2 | 2/1/3/5/4 | 1st |
| R.16 | 756 | 86 | 2 | 2/5/3/1/4 | 1st |
| R.17 | 643 | 118 | 2 | 3/2/1/5/4 | 2nd |
| R.18 | 604 | 120 | 2 | 2/5/3/1/4 | 1st |
| R.19 | 966 | 113 | 1 | 1/2/5/3/4 | 1st |
| R.20 | 694 | 141 | 2 | 1/2/4/3/5 | 2nd |
| R.21 | 718 | 209 | 2 | 2/4/5/1/3 | 1st |
| R.22 | 564 | 198 | 4 | 3/1/5/2/4 | 5th |
| R.23 | 790 | 92 | 1 | 1/2/5/3/4 | 1st |
| R.24 | 780 | 130 | 2 | 2/3/1/5/4 | 1st |
| R.25 | 771 | 108 | 2 | 4/2/3/5/1 | 2nd |
| R.26 | 657 | 155 | 2 | 2/4/1/5/3 | 1st |
| R.27 | 765 | 218 | 2 | 2/5/4/3/1 | 1st |
| R.28 | 651 | 285 | 2 | 2/5/3/1/4 | 1st |
| R.29 | 774 | 154 | 2 | 2/1/5/3/4 | 1st |
| R.30 | 683 | 125 | 2 | 2/4/3/5/1 | 1st |
| R.31 | 570 | 115 | 3 | 3/2/5/1/4 | 1st |
| R.32 | 527 | 121 | 4 | 4/1/2/3/5 | 1st |
| R.33 | 473 | 107 | 3 | 2/5/3/4/1 | 3rd |
| R.34 | 737 | 183 | 2 | 2/1/5/4/3 | 1st |
| R.35 | 473 | 107 | 3 | 4/2/5/3/1 | 4th |
| R.36 | 630 | 142 | 2 | 2/1/3/5/4 | 1st |
| R.37 | 608 | 120 | 2 | 2/4/3/1/5 | 1st |
| R.38 | 766 | 150 | 2 | 2/5/1/3/4 | 1st |
| R.39 | 821 | 164 | 1 | 1/2/3/5/4 | 1st |
| R.40 | 754 | 160 | 2 | 2/5/3/1/4 | 1st |

Table Column head definition:

- The pressure value is the average pen force in the digitizer.
- The speed value is the speed of the pen's pressure motion calculated by the digitizer.
- The labeling result column is the BFI labeling according to TABLE III. For example, data of respondent number R.12: the pressure value is 661 (medium), speed value is 107 (slow), and the BFI label is Agreeableness (2).
- The BFI test column is the result of manual BFI test from the respondent which is presented sequentially from the most dominant personality to the least dominant.
- The match column is the matching condition between the BFI label with the manual BFI test from the respondents. The definition of "1st" is the validation match with the first dominant personality and the definition of "2nd" is the validation match with the second dominant personality.

The percentage of the matching (see Table IV) shows 75% matching condition (30 of 40 matched). It shows that the personality of respondents could be predicted by using online signatures with decent accuracy. Although the validation accuracy is only 75%, this result was obtained only from the first predicted dominant personality (1st). When we combine with the second dominant personality BFI test (2nd), we found that the matching accuracy reached 90% (30 "1st " and 6 "2nd "). This result has shown the positive potency of applying digital signature for personality prediction.

The number of agreeableness personality reached 30 people out of 40 respondents, it shows that the agreeableness personality is dominant in this study. We think that one of the reasons is because in the BFI data label, the agreeableness personality has more portion of possibility than the other personality (based on Cattel's 16 Personality Factor).

From the 40 respondents, we found that there is no result indicated the openness personality. Our opinion for one of the reasons is because all of the respondents are technical students from electrical engineering, while openness personality tends to define people in more artistic personality or people with an artistic touch.

The best classification result using kNN is 87.5% of accuracy (k=5 and Percentage Split is 60%). This shows a quite good classification result of predicting personality using digitizer. However, if we use cross validation, the best classification result is only 77.7% (folds=4). Both results are not considered the imbalanced classes. For comparison, we use the Support Vector Machine (SVM) and Naive Bayes algorithms. The best accuracy for SVM and Naive Bayes is also 87.5% (Percentage Split is 60%).

For future work, the number of participants needs to be scale up, so the analysis regarding the precision of the system in predicting personality then can be optimized and overcome the imbalanced classes using Spread Sub Sample or Cost Sensitive Classifier. Other classifier for better accuracy result need also to be tested to find the best method for classification. For validation result, it is also open to analyze and validate the data using other method such as Enneagram so that we can understand better the personality prediction process using online signature.

## V. CONCLUSION

From the results we conclude that online signature has a high potential to be used as a personality prediction. However, a better and scale up study need to be performed, so that the comprehensiveness of the analysis then can be obtained in a better way. Furthermore, other validation methods for determining the personality manually such as Enneagram, need to be tested and compared, so that we can obtain a clearer understanding regarding the accuracy of implementing automatic personality prediction using an online signature. During our experiment we found that the portion of personality scoring tends to bring people into agreeableness personality. This finding certainly influences the whole result of this study. Another aspect that may also influence our result is the online signature device itself. Many studies shows that digital devices, however, give a sift of people's signature from its original version due to the process of digitizing. This issue is also an important aspect in determining the accuracy of the prediction. Even though from this study we also found that by using online signature the pressure and speed value tend to show a consistent pattern.

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